

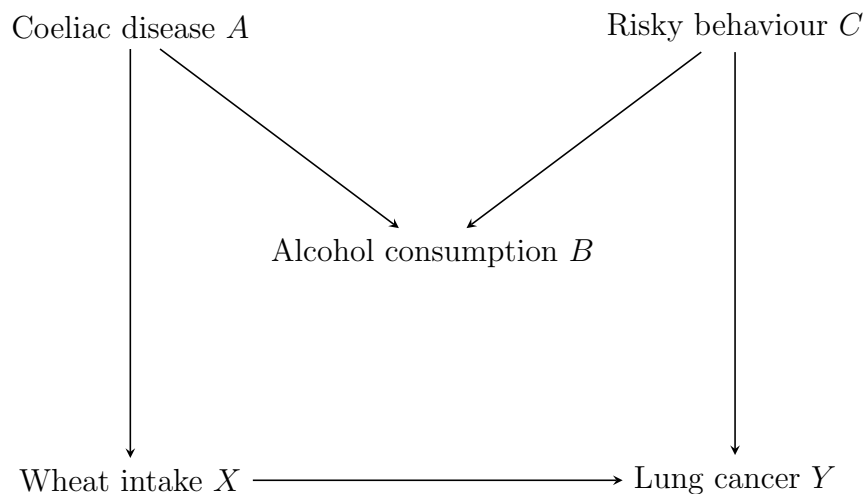
SMoDO - Practical 6 - Stata

Introduction

This practical is in two parts. In Part A you will use DAGs to inform selection of variables to controlled for in two example causal investigations. In Part B you will implement four variable selection methods used in investigations of causal effects, and compare the results: the full model approach, stepwise selection, change-in-estimates and double-selection.

Part A: DAGs

1. Consider the smoking and birth weight example from the lecture in Section 6.3.4, and the DAG in Figure 6.2. Suppose you wanted to estimate the causal effect of birth weight on survival. Which, if any, of the other variables in the DAG would you need to adjust for to do this?
2. In another study it is of interest to estimate the causal effect of wheat intake on risk of lung cancer. The DAG is shown below, and all variables depicted in the DAG are measured in the observed data. Would you need to adjust for alcohol consumption to estimate the causal effect of wheat intake on lung cancer?



Part B: Variable selection strategies

In this part we will use a data set containing information on the birth weight of 4642 babies, alongside information about the baby's mother and her maternal history, and some information about the baby's father. The aim is to estimate the causal effect of maternal smoking (yes or no) on the risk of low birth weight, defined as birth weight ≤ 2500 g. The data were described in a paper by Cattaneo 2010 (Journal of Econometrics 155: 138-154), and are available directly from Stata: use <https://www.stata-press.com/data/r16/cattaneo2>,

clear

1. Explore the dataset.
 - (a) Identify which variables correspond to the outcome Y , exposure X and set of potential confounders Z . Are there are variables you think should not be included in Z ?
 - (b) One issue to look for is where there are very few observations in a particular level of a categorical variable. If there are, we can merge categories to avoid running into problems during regression analysis. Are there any variables you would consider re-categorising in this data?
2. Summarise the exposure and outcome. Investigate the association between the mother's smoking status and risk of low birth weight, without adjustment for any other variables.
3. Fit an appropriate regression model for the outcome Y given X and the full set of potential confounders Z , and interpret the coefficient for X . What is the estimated marginal mean difference $E(Y|do(X = 1)) - E(Y|do(X = 0))$, and what assumptions do you make to obtain this? Interpret your results.
4. Perform forwards selection using a p-value for inclusion of $p^* = 0.1$. What set of adjustment variables do you end up with? Using this set of variables, use the standardisation method to obtain an estimate of the marginal mean difference, $E(Y|do(X = 1)) - E(Y|do(X = 0))$, and corresponding confidence interval.

For categorical variables we don't want each of the binary 'dummy' variables to be treated as separate variables in the selection process. They should be considered collectively. To force Stata to include or exclude all categories together you should enclose the variable in brackets in the command statement.

5. Perform the change-in-estimates method using the forwards selection approach. What set of adjustment variables do you end up with? Using this set of variables what is the estimated marginal mean difference, $E(Y|do(X = 1)) - E(Y|do(X = 0))$, and corresponding confidence interval.
6. Perform the double-selection approach. What set of adjustment variables do you end up with? Using this set of variables what is the estimated marginal mean difference, $E(Y|do(X = 1)) - E(Y|do(X = 0))$, and corresponding confidence interval. When using the `margins` command you can use the `vce(unconditional)` option to obtain standard errors and confidence intervals based on sandwich variance estimates.
7. Make a table presenting your results from the different methods and write an accompanying paragraph summarising the results. Include some explanation as to why you see different results from the different methods (if this is the case).
8. EXTRA: In Question 4, try backwards selection instead of forwards selection. Do you get a different set of variables? Try varying the p-value for inclusion/exclusion in the forwards/backwards selection methods to see what impact this has.